A1: Part 1 Extracting using SQL -Michel Ward

Part #1:

Step 1: Identifying the Client

See Appendix for the SQL code, results and key findings of this step.

Step 2: Creating the Client VIEW (invest.michel_ward)

SQL code with the Created VIEW with my name: invest.michel_ward. More details in Appendix

```sql
CREATE VIEW invest.michel_ward AS
SELECT DISTINCT s.major_asset_class, s.minor_asset_class,
       s.ticker, s.security_name, s.sec_type,
       p.date AS price_date, p.value AS price, p.price_type
FROM invest.customer_details c
LEFT JOIN invest.account_dim a
  ON c.customer_id = a.client_id
LEFT JOIN invest.holdings_current h
  ON a.account_id = h.account_id
LEFT JOIN invest.security_masterlist s
  ON h.ticker = s.ticker
LEFT JOIN invest.pricing_daily_new p
  ON s.ticker = p.ticker
WHERE c.customer_id = 128
  AND p.price_type = 'Adjusted'
  AND p.value IS NOT NULL
  AND p.value > 0;
```

Output

Action Output

#	Time	Action	Message
31	14:13:34	CREATE VIEW invest.michel_ward_prices AS SELECT DISTINCT s.major_asset_class, s.minor_asset_class, s.ticker, s.security_name, s.sec_type, p.date AS price_date, p.value AS price, p.price_type FROM in...	Error Code: 1050. Table 'michel_ward_prices' already exists
32	14:15:47	SELECT * FROM invest.michel_ward_prices LIMIT 0, 1000	1000 row(s) returned
33	14:17:53	SELECT DISTINCT s.major_asset_class, s.minor_asset_class, s.ticker, s.security_name, s.sec_type, p.date AS price_date, p.value AS price, p.price_type FROM invest.customer_details c LEFT JOIN invest.acc...	1000 row(s) returned
34	14:20:03	SELECT z.ticker, COUNT(*) AS n_rows, MIN(z.price_date) AS first_date, MAX(z.price_date) AS last_date FROM (SELECT DISTINCT s.ticker, p.date AS price_date, p.value AS price FROM invest.customer_detail...	27 row(s) returned
35	14:35:57	CREATE VIEW invest.michel_ward AS SELECT DISTINCT s.major_asset_class, s.minor_asset_class, s.ticker, s.security_name, s.sec_type, p.date AS price_date, p.value AS price, p.price_type FROM invest.cu...	0 row(s) affected

Figure 1: SQL code creating the VIEW invest.michel_ward

Question 1: Returns over 12, 24 and 36 Months

1.1 Method and SQL Code

The Return over a period of time uses the same formula the professor used in class:
(p1-p0)/p0

In this case:

- p1 = the price on the last date in the data, in this case, 2022-09-09
- p0 = the price at the start of the wanted period, it will be 12, 24 and 36 months

a. End price (p1)

To create the code, first we need to know where the data is. For that we use the FROM clause and I will use the VIEW that I created because it only has the data for Client ; Adjusted prices, so there are no extra filters to repeat

Then I will use SELECT ticker and price as p1

p1 is the price at the end of the period I want to analyze. In this code I want to know what is the price at the latest day (2022-09-09) so I can use the ROR formula.

```
SELECT ticker, price as p1
FROM invest.michel_ward
WHERE price_date = '2022-09-09';
```

b. Start price (p0)

Then we have to know what is the p0. I selected FROM my VIEW: invest.michel_ward, then WHERE date = the latest date – 12 months (or 24 and 36) using the DATE_SUB function

```
SELECT ticker, price as p0 FROM invest.michel_ward
WHERE price_date = DATE_SUB('2022-09-09', INTERVAL 12 MONTH);
```

c. ROR per security

I will combine the first and second p1 and p0 and add the ROR formula per security. For that I will have to use the column with ‘price’. I will also use a LEFT JOIN.

What the join does: it takes the same view twice and lines up the two dates side by side on one row. SQL calculates across columns in one row, so p0 and p1 must be placed side by side on the same row.

LEFT JOIN so that no ticker disappears from the results, even when it has no start price.

What the query is doing: it joins the view to itself.

- e (left side) = the end prices on 2022-09-09. That is all 27 tickers.
- s (right side) = the start prices on 2021-09-09. That is only 25 tickers, because CTA and CEG did not exist yet.

```
SELECT e.ticker, e.major_asset_class, e.minor_asset_class,

    s.price as p0, e.price as p1,
```

```
    (e.price - s.price)/s.price as ROR
FROM invest.michel_ward e
LEFT JOIN invest.michel_ward s
ON e.ticker = s.ticker
AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
WHERE e.price_date = '2022-09-09'
```

d. Market value per security

I need to add the market value per ticker. For that the market value = quantity*value. I will have to JOIN customer_details to account_dim to holdings_current. And filter by our customer id which equals to 128. The code is:

```
SELECT h.ticker, SUM(h.quantity * h.value) as mv
FROM invest.customer_details c
LEFT JOIN invest.account_dim a
ON c.customer_id = a.client_id
LEFT JOIN invest.holdings_current h
ON a.account_id = h.account_id
WHERE c.customer_id = 128
GROUP BY h.ticker
ORDER BY mv DESC;
```

After that I will combine this code with the previous codes:

```
SELECT e.ticker, e.major_asset_class, e.minor_asset_class,
    s.price as p0, e.price as p1,
    (e.price - s.price)/s.price as ROR,
    w.mv
FROM invest.michel_ward e
LEFT JOIN invest.michel_ward s
ON e.ticker = s.ticker
AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
```

```
LEFT JOIN (
 SELECT h.ticker, SUM(h.quantity * h.value) as mv
 FROM invest.customer_details c
 LEFT JOIN invest.account_dim a
 ON c.customer_id = a.client_id
 LEFT JOIN invest.holdings_current h
 ON a.account_id = h.account_id
 WHERE c.customer_id = 128
 GROUP BY h.ticker
) w
ON e.ticker = w.ticker
WHERE e.price_date = '2022-09-09'
ORDER BY ROR DESC;
```

This is the result I got for the 12 month window:

```sql
SELECT e.ticker, e.major_asset_class, e.minor_asset_class,
       s.price as p0, e.price as p1,
       (e.price - s.price)/s.price as ROR,
       w.mv
FROM invest.michel_ward e
LEFT JOIN invest.michel_ward s
ON e.ticker = s.ticker
AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
LEFT JOIN (
 SELECT h.ticker, SUM(h.quantity * h.value) as mv
 FROM invest.customer_details c
 LEFT JOIN invest.account_dim a
 ON c.customer_id = a.client_id
 LEFT JOIN invest.holdings_current h
 ON a.account_id = h.account_id
 WHERE c.customer_id = 128
 GROUP BY h.ticker
) w
ON e.ticker = w.ticker
WHERE e.price_date = '2022-09-09'
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ticker	major_asset_class	minor_asset_class	p0	p1	ROR	mv
ROK	equity	large_cap	309.596527	255.380005	-0.17511992955915806	252060.064935
TSCO	equity	large_cap	198.425659	200.919998	0.012570647428213892	170179.23830599998
CDNS	equity	large_cap	165.910004	174.679993	0.052859916753422595	164024.513427
BIDU	equity	small_cap	163.649994	138.429993	-0.15410939153471645	162793.671768
OTIS	equity	large_cap	89.267998	73.400002	-0.17775682613605834	60775.201656
DIS	equity	large_cap	185.910004	115.18	-0.3804529206507896	51370.280000000006
BIL	fixed_income		91.108643	91.470001	0.003966231831594677	48021.750524999996
GOOGL	equity	large_cap	143.541	110.650002	-0.2291400923777875	45477.150822
BNDX	fixed_income		55.547688	48.709999	-0.1230958343396758	40478.009169000004
DBC	commodities	precious_metals	19.110001	25.389999	0.3286236353415156	24932.979017999998
UNG	commodities	oil	17.49	27.889999	0.5946254431103489	23120.809171
LBAY	alternatives		22.668697	27.583	0.2167880668218379	22728.392
IAUM	commodities	precious_metals	17.958	17.139999	-0.045550785165385 84	16745.779023
DJP	commodities	precious_metals	27.790001	35.23	0.2677221566130924 5	14867.06
ARB	alternatives		25.801001	26.999001	0.046432307025607 28	14741.454546
GIGB	fixed income	corporate	53.870285	45.294998	-0.15918399169412234	14041.44938
CTA	alternatives		NULL	28.91	NULL	12864.95
CEG	equity	large_cap	NULL	87.510002	NULL	12776.460292
RLY	alternatives		25.376915	28.709999	0.13134315183701406	12632.39956
RINF	alternatives		29.311907	33.330002	0.13708064098320177	11432.190686
GLDM	commodities	precious_metals	35.700001	34.07	-0.04565829003758291	11277.17
FLSP	alternatives		20.386065	21.375	0.04851034272675975	9319.5
BCI	commodities	precious_metals	22.34663	27.42	0.22703065294409047	9295.380000000001
SGOL	commodities	precious_metals	17.25	16.450001	-0.04637675362318839	6826.750415
MSVX	alternatives		26.292395	24.25	-0.0776800668025868	6377.75
MARB	alternatives		19.639999	20.190001	0.028004176578624022	4300.470213
DBA	commodities	aggriculture	18.67	20.67	0.107123727905731 1	2315.04

Figure 2: 12M ROR and market value per security, sorted by market value

However, we also need the entire portfolio ROR and we need to set the parameters for all windows (12,24, and 36 months).

For that, this is what I did:

e. Portfolio ROR

After calculating the return of each security, I need the return of the whole portfolio. I cannot use a simple average of the 27 returns, because Client does not hold the same amount of money in each security. ROK is worth about $252K, while DBA is only about $2K. A loss in ROK should affect her portfolio much more than a loss in DBA. For that reason I use a weighted average, where each security's weight is its market value. This is the formula that I am going to use: Portfolio ROR = SUM(ROR × mv) / SUM(mv)
To build it, I use two subqueries and join them:

- Subquery r is my ROR query from step c. Here I change the LEFT JOIN to a plain JOIN. CTA and CEG have no start price, so their ROR is NULL, and a NULL return cannot be weighted. The plain JOIN removes them, so they don’t distort the result.
- Subquery w is my market value query from step d. It gives the dollar value of each ticker.

Because CTA and CEG were removed, I also calculate share_covered. This divides the market value included in the calculation by Client 128’s total portfolio value, and shows how much of her money the portfolio return represents.

```
SELECT SUM(r.ROR * w.mv)/SUM(w.mv) as portfolio_ROR,
      SUM(w.mv)/(SELECT SUM(h2.quantity * h2.value)
                 FROM invest.account_dim a2
                 JOIN invest.holdings_current h2
                 ON a2.account_id = h2.account_id
                 WHERE a2.client_id = 128) as share_covered
FROM (
 SELECT e.ticker, (e.price - s.price)/s.price as ROR
 FROM invest.michel_ward e
 JOIN invest.michel_ward s
 ON e.ticker = s.ticker
 AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
 WHERE e.price_date = '2022-09-09'
) r
JOIN (
 SELECT h.ticker, SUM(h.quantity * h.value) as mv
 FROM invest.customer_details c
 LEFT JOIN invest.account_dim a
 ON c.customer_id = a.client_id
 LEFT JOIN invest.holdings_current h
 ON a.account_id = h.account_id
 WHERE c.customer_id = 128
```

```
 GROUP BY h.ticker
) w
ON r.ticker = w.ticker;
```

1.2 Results: 12 Months

Per Security

```
SELECT e.ticker, e.major_asset_class, e.minor_asset_class,
       s.price as p0, e.price as p1,
       (e.price - s.price)/s.price as ROR,
       w.mv
FROM invest.michel_ward e
LEFT JOIN invest.michel_ward s
ON e.ticker = s.ticker
AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
LEFT JOIN (
 SELECT h.ticker, SUM(h.quantity * h.value) as mv
 FROM invest.customer_details c
 LEFT JOIN invest.account_dim a
 ON c.customer_id = a.client_id
 LEFT JOIN invest.holdings_current h
 ON a.account_id = h.account_id
 WHERE c.customer_id = 128
 GROUP BY h.ticker
) w
ON e.ticker = w.ticker
WHERE e.price_date = '2022-09-09';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ticker	major_asset_class	minor_asset_class	p0	p1	ROR	mv
LNG	commodities	oil	17.49	27.889999	0.5946254431103489	23120.809171
DBC	commodities	precious_metals	19.110001	25.389999	0.3286236353415156	24932.979017999998
DJP	commodities	precious_metals	27.790001	35.23	0.2677221566130924 5	14867.06
BCI	commodities	precious_metals	22.34663	27.42	0.2270306529440904 7	9295.380000000001
LBAY	alternatives		22.668697	27.583	0.2167880668218379	22728.392
RINF	alternatives		29.311907	33.330002	0.1370806409832017 7	11432.190686
RLY	alternatives		25.376915	28.709999	0.1313431518370140 6	12632.39956
DBA	commodities	aggriculture	18.67	20.67	0.1071237279057311	2315.04
CDNS	equity	large_cap	165.910004	174.679993	0.0528599167534225 95	164024.513427
FLSP	alternatives		20.386065	21.375	0.0485103427267597 5	9319.5
ARB	alternatives		25.801001	26.999001	0.0464323070256072 8	14741.454546
MARB	alternatives		19.639999	20.190001	0.0280041765786240 22	4300.470213
TSCO	equity	large_cap	198.425659	200.919998	0.0125706474282138 92	170179.23830599998
BIL	fixed_income		91.108643	91.470001	0.0039662318315946 77	48021.750524999996
IAUM	commodities	precious_metals	17.958	17.139999	-0.0455507851653858 4	16745.779023
GLDM	commodities	precious_metals	35.700001	34.07	-0.0456582900375829 1	11277.17
SGOL	commodities	precious_metals	17.25	16.450001	-0.0463767536231883 9	6826.750415
MSVX	alternatives		26.292395	24.25	-0.0776800668025868	6377.75
BNDX	fixed_income		55.547688	48.709999	-0.1230958343967580	40478.009169000004
BIDU	equity	small_cap	163.649994	138.429993	-0.1541093915347164 5	162793.671768
GIGB	fixed income	corporate	53.870285	45.294998	-0.1591839916941223 4	14041.44938
ROK	equity	large_cap	309.596527	255.380005	-0.1751199295591580 6	252060.064935
OTIS	equity	large_cap	89.267998	73.400002	-0.1777568261360583 4	60775.201656
GOOGL	equity	large_cap	143.541	110.650002	-0.2291400923777875	45477.150822
DIS	equity	large_cap	185.910004	115.18	-0.3804529206507896	51370.280000000006
CTA	alternatives		NULL	28.91	NULL	12864.95
CEG	equity	large_cap	NULL	87.510002	NULL	12776.460292

Figure 3: 12M ROR and market value per security, sorted by ROR

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Results by asset class (weighted by mv, calculated from this output)

Asset class	Market value	12M return	$ impact
Equity (excl. CEG)	$907K	−10.9%	−$99K
Commodities	$109K	+24.4%	+$27K

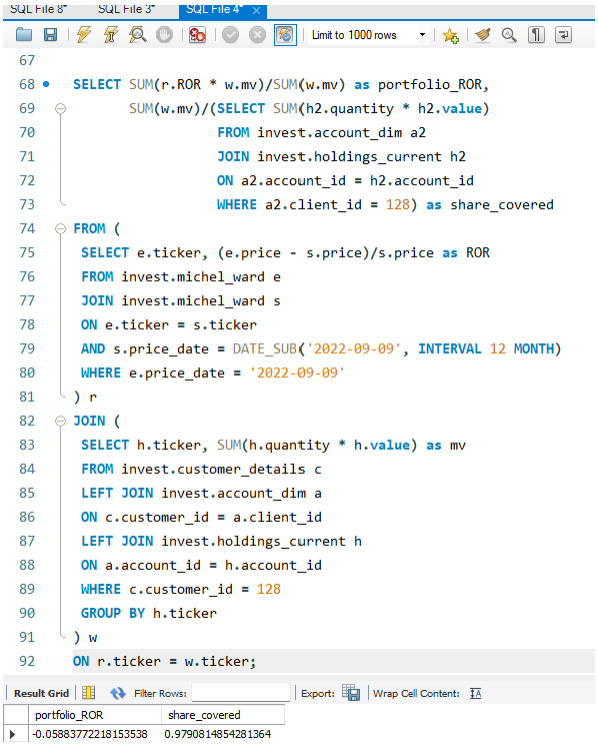
Asset class	Market value	12M return	$ impact
Alternatives (excl. CTA)	$82K	+10.9%	+$9K
Fixed income	$103K	−6.9%	−$7K

- Equity drove the loss. About 75% of her money is in stocks, and they lost about $99K. Everything else combined added back only about $29K.
- Concentration hurt. ROK, her largest position at about 21% of the entire portfolio, lost 17.5%, or about −$44K. That single stock caused over half the total loss.
- Diversifiers worked, but were too small. Commodities returned +24% and alternatives +11%, the best performing classes. But together they are only about 16% of the portfolio, so they couldn't offset equity.
- Bonds did not protect her. BNDX (−12.3%) and GIGB (−15.9%) fell as interest rates rose in 2022. Only BIL, short term T bills, held its value (+0.4%). Her "safe" layer was not safe this year.

Questions:

1. Do the 24M and 36M windows tell the same story, or was this a bad year after strong gains?
2. How did a benchmark such as SPY or AGG perform over the same 12 months? Did she lose more or less than the market?

Whole Portfolio



```sql
SELECT SUM(r.ROR * w.mv)/SUM(w.mv) as portfolio_ROR,
       SUM(w.mv)/(SELECT SUM(h2.quantity * h2.value)
                  FROM invest.account_dim a2
                  JOIN invest.holdings_current h2
                  ON a2.account_id = h2.account_id
                  WHERE a2.client_id = 128) as share_covered
FROM (
 SELECT e.ticker, (e.price - s.price)/s.price as ROR
 FROM invest.michel_ward e
 JOIN invest.michel_ward s
 ON e.ticker = s.ticker
 AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
 WHERE e.price_date = '2022-09-09'
 ) r
JOIN (
 SELECT h.ticker, SUM(h.quantity * h.value) as mv
 FROM invest.customer_details c
 LEFT JOIN invest.account_dim a
 ON c.customer_id = a.client_id
 LEFT JOIN invest.holdings_current h
 ON a.account_id = h.account_id
 WHERE c.customer_id = 128
 GROUP BY h.ticker
 ) w
 ON r.ticker = w.ticker;
```

portfolio_ROR	share_covered
-0.05883772218153538	0.9790814854281364

Figure 4: 12M portfolio ROR and share covered

Business Insights

- She lost money over the year. For every $100 invested on 2021-09-09, she had about $94.12 on 2022-09-09.
- The result is reliable. With 97.9% coverage, the two excluded funds are too small to change the conclusion.
- The loss is smaller than the equity loss. Equity alone fell about 10.9%. Commodities and alternatives pulled the portfolio up to −5.9%, so diversification helped, just not enough.

Questions:

1. Is the 12M loss part of a longer trend, or a single bad year after strong 24M and 36M returns?
2. Does this level of loss fit her risk tolerance and time horizon?

1.3 Results: 24 Months

```
93
94 •  SELECT e.ticker, e.major_asset_class, e.minor_asset_class,
95           s.price as p0, e.price as p1,
96           (e.price - s.price)/s.price as ROR,
97           w.mv
98    FROM invest.michel_ward e
99    LEFT JOIN invest.michel_ward s
100   ON e.ticker = s.ticker
101   AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 24 MONTH)
102   LEFT JOIN (
103    SELECT h.ticker, SUM(h.quantity * h.value) as mv
104    FROM invest.customer_details c
105    LEFT JOIN invest.account_dim a
106    ON c.customer_id = a.client_id
107    LEFT JOIN invest.holdings_current h
108    ON a.account_id = h.account_id
109    WHERE c.customer_id = 128
110    GROUP BY h.ticker
111   ) w
112   ON e.ticker = w.ticker
113   WHERE e.price_date = '2022-09-09';
114
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ticker	major_asset_class	minor_asset_class	p0	p1	ROR	mv
UNG	commodities	oil	12.79	27.889999	1.1806097732603598	23120.809171
DBC	commodities	precious_metals	12.94	25.389999	0.9621328438948996	24932.979017999998
DJP	commodities	precious_metals	19.799999	35.23	0.779293019156213	14867.06
CDNS	equity	large_cap	104.339996	174.679993	0.6741422244256171	164024.513427
BCI	commodities	precious_metals	16.665043	27.42	0.6453602909995492	9295.380000000001
RLY	alternatives		19.383696	28.709999	0.4811416254155038	12632.39956
TSCO	equity	large_cap	137.691055	200.919998	0.45920879174032025	170179.23830599998
GOOGL	equity	large_cap	77.361504	110.650002	0.4302979683538728	45477.150822
DBA	commodities	aggriculture	14.62	20.67	0.4138166894664845	2315.04
RINF	alternatives		25.024395	33.330002	0.3319004115783819	11432.190686
OTIS	equity	large_cap	60.528042	73.400002	0.21266110012281583	60775.201656
ROK	equity	large_cap	217.124191	255.380005	0.17619323680059223	252060.064935
BIDU	equity	small_cap	119.400002	138.429993	0.15938015645929382	162793.671768
ARB	alternatives		24.705336	26.999001	0.09284087453819696	14741.454546
MARB	alternatives		19.370001	20.190001	0.04233350323523475	4300.470213
MSVX	alternatives		23.505405	24.25	0.03167760776723483	6377.75
FLSP	alternatives		20.766087	21.375	0.029322471778144874	9319.5
BIL	fixed_income		91.185356	91.470001	0.003121608693395874	48021.750524999996
BNDX	fixed_income		55.252884	48.709999	-0.11841707665431542	40478.009169000004
GLDM	commodities	precious_metals	38.82	34.07	-0.12235960844925296	11277.17
SGOL	commodities	precious_metals	18.75	16.450001	-0.12266661333333331	6826.750415
DIS	equity	large_cap	133.360001	115.18	-0.1363227419291936	51370.280000000006
GIGB	fixed income	corporate	52.704945	45.294998	-0.14059301266702778	14041.44938
LBAY	alternatives		NULL	27.583	NULL	22728.392
CTA	alternatives		NULL	28.91	NULL	12864.95
IAUM	commodities	precious_metals	NULL	17.139999	NULL	16745.779023
CEG	equity	large_cap	NULL	87.510002	NULL	12776.460292

Figure 5: 24M ROR and market value per security

Results by asset class (weighted by mv, calculated from this output)

Asset class	Market value	24M return	$ impact
Equity (excl. CEG)	$907K	+31.4%	+$285K
Commodities (excl. IAUM)	$93K	+73.0%	+$68K
Alternatives (excl. LBAY, CTA)	$59K	+20.2%	+$12K

Asset class	Market value	24M return	$ impact
Fixed income	$103K	−6.5%	−$7K

Business insights

- Two years tell a very different story than one. Over 24 months, most of her holdings gained. The loss came entirely in the last 12 months, so she gave back part of a very strong first year.
- CDNS was the star. It returned +67.4%, adding about $111K, the largest contribution of any holding.
- ROK peaked and fell. It is up 17.6% over 24M but down 17.5% over 12M, so most of its gains disappeared in the last year. GOOGL (+43.0% vs −22.9%) and BIDU (+15.9% vs −15.4%) show the same pattern.
- DIS is weak in both windows (−13.6% at 24M, −38.0% at 12M). It is the only stock losing over both periods.
- Commodities were exceptional, with UNG at +118%, DBC +96% and DJP +78%. But they are only about 8% of the portfolio.

Whole Portfolio

```
SELECT SUM(r.ROR * w.mv)/SUM(w.mv) as portfolio_ROR,
       SUM(w.mv)/(SELECT SUM(h2.quantity * h2.value)
                  FROM invest.account_dim a2
                  JOIN invest.holdings_current h2
                  ON a2.account_id = h2.account_id
                  WHERE a2.client_id = 128) as share_covered
FROM (
 SELECT e.ticker, (e.price - s.price)/s.price as ROR
 FROM invest.michel_ward e
 JOIN invest.michel_ward s
 ON e.ticker = s.ticker
 AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 24 MONTH)
 WHERE e.price_date = '2022-09-09'
) r
JOIN (
 SELECT h.ticker, SUM(h.quantity * h.value) as mv
 FROM invest.customer_details c
 LEFT JOIN invest.account_dim a
 ON c.customer_id = a.client_id
 LEFT JOIN invest.holdings_current h
 ON a.account_id = h.account_id
 WHERE c.customer_id = 128
 GROUP BY h.ticker
) w
ON r.ticker = w.ticker;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

portfolio_ROR	share_covered
0.30798242789504315	0.946878068675569

Figure 6: 24M portfolio ROR and share covered

Business Insights:

- Strong two year result. $100 invested on 2020-09-09 was worth about $131 on 2022-09-09.
- The gain came early and was partly given back. The 12M return was −5.9%, so almost all of the growth happened in the first year (September 2020 to September 2021), during the post COVID market recovery. The second year erased part of it.

Questions:

1. Does the 36M window confirm a long term gain, or does it show a pattern of high swings?
2. Was the first year's gain mainly the 2020 and 2021 post COVID recovery? If so, is that repeatable?

1.4 Results: 36 Months

Per Security

```
SELECT e.ticker, e.major_asset_class, e.minor_asset_class,
       s.price as p0, e.price as p1,
       (e.price - s.price)/s.price as ROR,
       w.mv
FROM invest.michel_ward e
LEFT JOIN invest.michel_ward s
ON e.ticker = s.ticker
AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 36 MONTH)
LEFT JOIN (
 SELECT h.ticker, SUM(h.quantity * h.value) as mv
 FROM invest.customer_details c
 LEFT JOIN invest.account_dim a
 ON c.customer_id = a.client_id
 LEFT JOIN invest.holdings_current h
 ON a.account_id = h.account_id
 WHERE c.customer_id = 128
 GROUP BY h.ticker
) w
ON e.ticker = w.ticker
WHERE e.price_date = '2022-09-09';
```

ticker	major_asset_class	minor_asset_class	p0	p1	ROR	mv
CDNS	equity	large_cap	68.410004	174.679993	1.5534276097981223	164024.513427
TSCO	equity	large_cap	98.472733	200.919998	1.0403617917145582	170179.23830999998
GOOGL	equity	large_cap	60.3635	110.650002	0.8361031470127025	45477.150822
DBC	commodities	precious_metals	15.019402	25.389999	0.6904800204428911	24932.979017999998
ROK	equity	large_cap	153.433105	255.380005	0.6644087467750196	252060.064935
DJP	commodities	precious_metals	21.969999	35.23	0.6035503688461704	14867.06
BCI	commodities	precious_metals	18.055443	27.42	0.5186556209116554	9295.380000000001
DBA	commodities	aggriculture	14.610525	20.67	0.4147335560237367	2315.04
RLY	alternatives		20.777113	28.709999	0.3818088682484424	12832.39956
RINF	alternatives		24.292809	33.330002	0.3720110342181153	11432.190686
BIDU	equity	small_cap	105.059998	138.429993	0.3176279805373688	162793.671768
UNG	commodities	oil	22.48	27.889999	0.2406383185053803	23120.809171
SGOL	commodities	precious_metals	14.449	16.450001	0.1384871617412970 2	6826.750415
GLDM	commodities	precious_metals	29.940001	34.07	0.1379425133762775 14	11277.17
BIL	fixed_income		90.353134	91.470001	0.0123611318230533 02	48021.750524999996
GIGB	fixed income	corporate	48.367916	45.294998	-0.0635321563161828 5	14041.44938
BNDX	fixed_income		54.180943	48.709999	-0.1009754296819823 9	40478.009169000004
DIS	equity	large_cap	138.003189	115.18	-0.1653816059783037 3	51370.280000000006
MARB	alternatives		[redacted]	20.190001	[redacted]	4300.470213
FLSP	alternatives		[redacted]	21.375	[redacted]	9319.5
LBAY	alternatives		[redacted]	27.583	[redacted]	22728.392
CTA	alternatives		[redacted]	28.91	[redacted]	12864.95
IAUM	commodities	precious_metals	[redacted]	17.139999	[redacted]	16745.779023
MSVX	alternatives		[redacted]	24.25	[redacted]	6377.75
CEG	equity	large_cap	[redacted]	87.510002	[redacted]	12776.460292
OTIS	equity	large_cap	[redacted]	73.400002	[redacted]	60775.201656
ARB	alternatives		[redacted]	26.999001	[redacted]	14741.454946

Figure 7: 36M ROR and market value per security

Whole Portfolio

```sql
SELECT SUM(r.ROR * w.mv)/SUM(w.mv) as portfolio_ROR,
       SUM(w.mv)/(SELECT SUM(h2.quantity * h2.value)
                  FROM invest.account_dim a2
                  JOIN invest.holdings_current h2
                  ON a2.account_id = h2.account_id
                  WHERE a2.client_id = 128) as share_covered
FROM (
  SELECT e.ticker, (e.price - s.price)/s.price as ROR
  FROM invest.michel_ward e
  JOIN invest.michel_ward s
  ON e.ticker = s.ticker
  AND s.price_date = DATE_SUB('2022-09-09', INTERVAL 36 MONTH)
  WHERE e.price_date = '2022-09-09'
 ) r
JOIN (
  SELECT h.ticker, SUM(h.quantity * h.value) as mv
  FROM invest.customer_details c
  LEFT JOIN invest.account_dim a
  ON c.customer_id = a.client_id
  LEFT JOIN invest.holdings_current h
  ON a.account_id = h.account_id
  WHERE c.customer_id = 128
  GROUP BY h.ticker
 ) w
 ON r.ticker = w.ticker;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

portfolio_ROR	share_covered
0.6809281837577041	0.8689565014877072

Figure 8: 36M portfolio ROR and share covered

Results by Asset Class

Asset class	Market value	36M return	$ impact
Equity (excl. CEG, OTIS)	$846K	+80.5%	+$681K
Commodities (excl. IAUM)	$93K	+43.2%	+$40K
Alternatives (only RLY, RINF)	$24K	+37.7%	+$9K
Fixed income	$103K	−4.3%	−$4K

- A few stocks built her wealth. CDNS alone added about $255K, roughly 35% of the total gain. CDNS, TSCO, ROK and BIDU together made about 90% of it.
- Equity was the engine. Stocks produced about 94% of the 36M gain. The same concentration that created the gains also caused the 12M loss.
- Gold behaved differently over three years. SGOL and GLDM gained about 14% over 36M but lost money over 24M and 12M. Gold rose during the 2020 crisis, then gave the gains back.

1.5 Summary Across Windows

Window	Portfolio ROR	Coverage
12M	−5.9%	97.9%
24M	+30.8%	94.7%
36M	+68.1%	86.9%

Questions to explore next

1. Should she lock in some CDNS and TSCO gains before momentum fades further?
2. Should ROK, her largest position, be reduced now that its gains are disappearing?
3. Should DIS, BNDX and GIGB be sold or replaced, since they lost in every window?
4. For the client: does she want to protect the gains she has made, or keep chasing growth with high concentration?

I attempted a stored procedure with the window as an input; due to a Workbench delimiter error, I ran each window separately by changing the INTERVAL value.

Question 2: 12 Month Risk (sigma) and Average Daily Return (mu)

2.1 Method and SQL Code

This is the same structure as my class code “mu and sigma per ticker, using LAG”. I just switch it to my view and my 12M window.
Q2 needs every daily return in the year, so each day is compared with the day before: daily ROR = (p1 − p0) / p0, where p1 is today’s price and p0 is yesterday’s.

a. p1 and p0 (LAG)
p0 and p1 must sit on the same row. LAG(price, 1) pulls the previous row’s price forward.
Then FROM my VIEW, WHERE the date is within the last 12 months and SELECT ticker, date, price as p1, and LAG as p0

```
SELECT ticker, price_date, price as p1,

    LAG(price, 1) OVER(
```

```
       PARTITION BY ticker
       ORDER BY price_date
      ) as p0
FROM invest.michel_ward
WHERE price_date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH);
```

b. Daily ROR

I wrapped first step in parentheses as subquery z, the same as in class. Then SELECT (z.p1 – z.p0)/z.p0 as ROR

```
SELECT z.ticker, z.price_date, z.p1, z.p0,
      (z.p1 - z.p0)/z.p0 as ROR
FROM (
 SELECT ticker, price_date, price as p1,
       LAG(price, 1) OVER(
        PARTITION BY ticker
        ORDER BY price_date
       ) as p0
 FROM invest.michel_ward
 WHERE price_date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
) z;
```

c. mu and sigma per security

1. I wrapped second step as a subquery again
2. **WHERE** o.ROR IS NOT NULL, which removes the first day of each ticker. This is the same filter as the class Dubai query.
3. **GROUP BY** ticker, giving one result per security.
4. SELECT AVG(o.ROR) as mu, STD(o.ROR) as sigma, COUNT(o.ROR) as n_days

```
SELECT o.ticker, o.major_asset_class,
```

```
      AVG(o.ROR) as mu,
      STD(o.ROR) as sigma,
      COUNT(o.ROR) as n_days
FROM (
 SELECT z.ticker, z.major_asset_class,
       (z.p1 - z.p0)/z.p0 as ROR
 FROM (
  SELECT ticker, major_asset_class, price_date, price as p1,
        LAG(price, 1) OVER(
         PARTITION BY ticker
         ORDER BY price_date
        ) as p0
  FROM invest.michel_ward
  WHERE price_date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
 ) z
) o
WHERE o.ROR IS NOT NULL
GROUP BY o.ticker, o.major_asset_class
```

2.2 Results, Key Findings and Business Insights

```sql
SELECT o.ticker, o.major_asset_class,
       AVG(o.ROR) as mu,
       STD(o.ROR) as sigma,
       COUNT(o.ROR) as n_days
FROM (
 SELECT z.ticker, z.major_asset_class,
        (z.p1 - z.p0)/z.p0 as ROR
 FROM (
  SELECT ticker, major_asset_class, price_date, price as p1,
         LAG(price, 1) OVER(
          PARTITION BY ticker
          ORDER BY price_date
         ) as p0
  FROM invest.michel_ward
  WHERE price_date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
 ) z
) o
WHERE o.ROR IS NOT NULL
GROUP BY o.ticker, o.major_asset_class
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ticker	major_asset_class	mu	sigma	n_days
UNG	commodities	0.0029565260391413075	0.04679267081824431	252
BIDU	equity	0.0001752242439155321	0.04247624560846277	252
CEG	equity	0.005142053261819878	0.032644097373273617	161
CDNS	equity	0.0005172889863138617	0.024997443563370716	252
GOOGL	equity	-0.00080258055372721 52	0.021452694892630023	252
ROK	equity	-0.0005383799828615856	0.021079427038758494	252
TSCO	equity	0.000271200643942817	0.020871363013411 62	252
DIS	equity	-0.0017023671033584713	0.019747893239064 13	252
OTIS	equity	-0.0006413008556868329	0.016422797450845208	252
DBC	commodities	0.0012618309005981642	0.016260232639825532	252
DJP	commodities	0.001074683096842222	0.016233167163001758	252
BCI	commodities	0.0009208222153526999	0.014678730333099809	252
RINF	alternatives	0.0005850210768678907	0.012269567696105453	252
RLY	alternatives	0.0005443047923744114	0.010416942673669724	252
DBA	commodities	0.0004515499755440898	0.009755902331671966	252
CTA	alternatives	0.00109309531675919	0.009256101599352083	128
LBAY	alternatives	0.0008198413664641813	0.009042190163673166	252
GLDM	commodities	-0.0001482051413082971	0.008620259927985055	252
FLSP	alternatives	0.0002251834320612 2965	0.008607266950123893	252
SGOL	commodities	-0.0001513381123289644 6	0.008604188810403273	252
IAUM	commodities	-0.0001480905042886996 7	0.008582079344920663	252
GIGB	fixed income	-0.0006729189855169743	0.005448658225924101	252
MSVX	alternatives	-0.0003080103844002208 6	0.005065712555844975	252
ARB	alternatives	0.0001880738250165901	0.003990094643374493	252
BNDX	fixed_income	-0.0005141802762166533	0.003726292513735451	252
MARB	alternatives	0.0001129148382999403 4	0.002572550406795592 3	252
BIL	fixed_income	0.0000157150389826717 5	0.000118580160150265 79	252

Figure 9: 12M mu, sigma and trading days per security

After reading Key Findings:

Risk by asset class (sigma range)

Asset class	Lowest sigma	Highest sigma
Equity	1.64% (OTIS)	4.24% (BIDU)
Commodities	0.86% (gold ETFs)	4.68% (UNG)
Alternatives	0.26% (MARB)	1.23% (RINF)
Fixed income	0.01% (BIL)	0.55% (GIGB)

Business insights

- Her biggest positions are among her riskiest. BIDU (#2), CDNS (#4), ROK (#6) and TSCO (#7) are all in the top 8 for sigma, and together they hold about 61% of her money. Most of her wealth sits in the most volatile part of the portfolio.
- ROK takes risk without reward. It is her largest position (about 21%), with a sigma of 2.11% and a negative mu (−0.05% per day). She is exposed to large daily swings in a stock that lost money on average.
- DIS is the clearest example of bad risk. It has the worst mu (−0.17% per day) with a high sigma (1.97%). That matches Q1, where it lost in every window.

2.3 Questions and Recommendations

Questions

1. Which securities give the best return per unit of risk?
2. What is the sigma of the whole portfolio? Diversification should make it lower than most individual stocks.

Recommendations

1. Trim ROK and BIDU. They are large positions with high risk and poor recent returns. Bring each closer to 5 to 10% of the portfolio.
2. Sell or sharply reduce DIS. It has high risk, the worst daily return, and losses in every Q1 window.
3. Keep UNG small. If she keeps it, treat it as a small tactical position, under 2%, given its extreme volatility.

Question 3: Adding a New Investment

3.1 Choosing the Asset Type

To answer this question, I want to know what types of investments are there. For that I am going to check FROM invest.security_masterlist, what are the major assets and minor assets available:

```
208 •  SELECT major_asset_class, minor_asset_class, COUNT(DISTINCT ticker) as n_tickers
209    FROM invest.security_masterlist
210    GROUP BY major_asset_class, minor_asset_class
211    ORDER BY major_asset_class, minor_asset_class;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

major_asset_class	minor_asset_class	n_tickers
alternatives		23
commodities	aggriculture	1
commodities	cannabis	7
commodities	carbon	1
commodities	gold	2
commodities	oil	5
commodities	precious_metals	14
commodities	silver	1
equity		1
equity	emerging_markets	1
equity	large_cap	364
equity	small_cap	4
equty	emerging_markets	1
fixed income	corporate	1
fixed income	inv_grade_corp	1
fixed income	mid-term-corporate	1
fixed income corp...		1
fixed_income		24
fixed_income	emerging_markets	1

Figure 10: Number of tickers by major and minor asset class

My suggestion: fixed income

With only about 29 tickers, fixed income is easy to review, and it fixes her weakest area: bonds that lost money and a thin defensive layer. Equity has 364+ individual stocks, too many to compare, and adding another single stock would increase concentration instead of reducing it. I could add a broad fund like SPY, but I will stick with fixed income.

Another thing I need to know here is my client's risk tolerance. From the data I collected, it seems like she likes riskier investments. However, her current allocation may be the result of growth that was never rebalanced rather than a deliberate choice. Since no information from her confirms her risk tolerance, I will follow the portfolio management rule of reducing risk. As discussed in class, wealth management clients look for the sweet spot of steady returns slightly above inflation, while keeping risk and taxes low. Before implementing this change, I would confirm Client ; goals, time horizon and risk tolerance.

3.2 Screening Fixed Income Securities

Reasoning:

FROM security_masterlist. This is the table that lists every security and its classification, so it is the starting point for finding new investments.

LEFT JOIN pricing_daily_new. This attaches each security's price history. LEFT keeps securities with no prices, so nothing is hidden.

Filters inside ON (price_type = 'Adjusted' and value IS NOT NULL). Only real Adjusted prices are counted, the same rules as my view. They are placed in ON, not WHERE, so securities without prices still appear.

WHERE major_asset_class LIKE ‘fixed%’. The class is spelled three ways in the database: fixed_income, fixed income and fixed income corporate. LIKE 'fixed%' catches every spelling, because % means “any text after fixed”. An exact match such as = 'fixed_income' would miss some bonds.

COUNT, MIN and MAX. These show how much price history each security has. A new investment needs at least 12 months of prices (from 2021-09-09) to calculate its sigma the same way as in Q2.

GROUP BY. This gives one row per security

```
213 •  SELECT s.ticker, s.security_name, s.major_asset_class, s.minor_asset_class,
214           COUNT(p.value) as n_days,
215           MIN(p.date) as first_date,
216           MAX(p.date) as last_date
217    FROM invest.security_masterlist s
218    LEFT JOIN invest.pricing_daily_new p
219    ON s.ticker = p.ticker
220    AND p.price_type = 'Adjusted'
221    AND p.value IS NOT NULL
222    WHERE s.major_asset_class LIKE 'fixed%'
223    GROUP BY s.ticker, s.security_name, s.major_asset_class, s.minor_asset_class
224    ORDER BY s.major_asset_class, s.ticker;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ticker	security_name	major_asset_class	minor_asset_class	n_days	first_date	last_date
GIGB	Goldman Sachs Acss Invmt Grd Corp Bd ETF	fixed income	corporate	1324	2017-06-08	2022-09-09
LQD	iShares iBoxx Invmt Grade Corp Bd ETF	fixed income	inv_grade_corp	3950	2007-01-03	2022-09-09
SCHI	Shwab 5-10 year corporate bond	fixed income	mid-term-corporate	734	2019-10-11	2022-09-09
USIG	iShares Broad USD Invm Grd Corp Bd ETF	fixed income corporate		3944	2007-01-11	2022-09-09
AGG	iShares Core U.S. Aggregate Bond ETF	fixed_income		3950	2007-01-03	2022-09-09
BIL	SPDR Bloomberg 1-3 Month T-Bill ETF	fixed_income		3849	2007-05-30	2022-09-09
BND	Vanguard Total Bond Market ETF	fixed_income		3884	2007-04-10	2022-09-09
BNDX	Vanguard Total International Bond ETF	fixed_income		2335	2013-06-04	2022-09-09
BSV	Vanguard Short-Term Bond ETF	fixed_income		3884	2007-04-10	2022-09-09
EMB	iShares J.P. Morgan USD Emerging Markets Bon…	fixed_income		3707	2007-12-19	2022-09-09
GOVT	iShares U.S. Treasury Bond ETF	fixed_income		2654	2012-02-24	2022-09-09
IEF	iShares 7-10 Year Treasury Bond ETF	fixed_income		3950	2007-01-03	2022-09-09
IGSB	iShares 1-5 Year Investment Grade Corporate B…	fixed_income		3944	2007-01-11	2022-09-09
IUSB	iShares Core Total USD Bond Market ETF	fixed_income		2077	2014-06-12	2022-09-09
JPST	JPMorgan Ultra-Short Income ETF	fixed_income		1337	2017-05-19	2022-09-09
MBB	iShares MBS ETF	fixed_income		3900	2007-03-16	2022-09-09
MUB	iShares National Muni Bond ETF	fixed_income		3778	2007-09-10	2022-09-09
SCHP	Schwab U.S. TIPS ETF	fixed_income		3046	2010-08-05	2022-09-09
SHV	iShares Short Treasury Bond ETF	fixed_income		3944	2007-01-11	2022-09-09
SHY	iShares 1-3 Year Treasury Bond ETF	fixed_income		3950	2007-01-03	2022-09-09
TIP	iShares TIPS Bond ETF	fixed_income		3950	2007-01-03	2022-09-09
TLT	iShares 20+ Year Treasury Bond ETF	fixed_income		3950	2007-01-03	2022-09-09
VCIT	Vanguard Intermediate-Term Corporate Bond ETF	fixed_income		3221	2009-11-23	2022-09-09
VCSH	Vanguard Short-Term Corporate Bond ETF	fixed_income		3221	2009-11-23	2022-09-09
VGSH	Vanguard Short-Term Treasury ETF	fixed_income		3221	2009-11-23	2022-09-09
VMBS	Vanguard Mortgage-Backed Securities ETF	fixed_income		3221	2009-11-23	2022-09-09
VTEB	Vanguard Tax-Exempt Bond ETF	fixed_income		1774	2015-08-25	2022-09-09
VTIP	Vanguard Short-Term Inflation-Protected Securi…	fixed_income		2491	2012-10-16	2022-09-09
VWO	Vanguard Ftse Emerging Markets	fixed_income	emerging_markets	3950	2007-01-03	2022-09-09

Figure 11: Fixed income securities and price history

The fixed income query returned 29 securities. I removed the three Client already owns (BIL, BNDX and GIGB), because the question asks for a new investment. I also removed VWO. Although the database classifies it as fixed income, it is the Vanguard FTSE Emerging Markets fund, which is an emerging markets stock fund. This is another data quality issue in the security masterlist.

From the remaining funds, I excluded the ones that would repeat the problems I found in Questions 1 and 2:

- Long term bonds (TLT, IEF, LQD, VCIT). These are very sensitive to rising interest rates, the same reason BNDX and GIGB lost money.
- Emerging market bonds (EMB). These add country risk, and Client already has China exposure through BIDU.
- Municipal bonds (MUB, VTEB). Their tax advantage only applies to US taxpayers, and Client ’s tax situation is unknown.

I selected this shortlist:

Ticker	Fund	Why it fits
VTIP	Vanguard Short Term Inflation Protected Securities	Protects against inflation, with short duration and low rate risk
IGSB	iShares 1 to 5 Year Investment Grade Corporate Bond	Short term corporate bonds with a bit more yield than Treasuries
VCSH	Vanguard Short Term Corporate Bond	Short term corporate bonds with a bit more yield than Treasuries
SHY	iShares 1 to 3 Year Treasury Bond	Very safe, with more return potential than BIL
TIP	iShares TIPS Bond	Inflation protection across all maturities
AGG	iShares Core US Aggregate Bond	Broad bond market benchmark, also used in class

My first choice is **VTIP**, because it addresses the main risk of 2022, inflation, while staying short term like BIL, which held its value when rates rose. However, I will not decide on description alone. Using the same method as Question 2, I will calculate the average daily return (mu) and risk (sigma) of each candidate over the most recent 12 months. The best fit will be the fund with low sigma and the strongest mu, because it should lower Client ; portfolio risk without giving up return.

3.3 Comparing the Candidates (mu and sigma)

To compare the candidates, I used the same mu and sigma method as in Question 2, applied directly to the pricing_daily_new table, because these funds are not part of Client ; view. First, I filtered Adjusted prices for the six candidates over the most recent 12 months. Then I used LAG to place each day’s previous price next to the current price, calculated the daily return with (p1 − p0)/p0, and grouped by ticker to get the average daily return (mu) and the risk (sigma). This makes the candidates directly comparable with her current holdings.

```sql
SELECT o.ticker,
       AVG(o.ROR) as mu,
       STD(o.ROR) as sigma,
       COUNT(o.ROR) as n_days
FROM (
 SELECT z.ticker, (z.p1 - z.p0)/z.p0 as ROR
 FROM (
  SELECT ticker, date, value as p1,
        LAG(value, 1) OVER(
         PARTITION BY ticker
         ORDER BY date
        ) as p0
  FROM invest.pricing_daily_new
  WHERE price_type = 'Adjusted'
  AND ticker IN ('VTIP','IGSB','VCSH','SHY','TIP','AGG')
  AND date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
 ) z
) o
WHERE o.ROR IS NOT NULL
GROUP BY o.ticker
ORDER BY sigma;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ticker	mu	sigma	n_days
SHY	-0.00016922169484753574	0.0012677957835592769	252
VCSH	-0.00025900390210163335	0.0020747811403880 46	252
IGSB	-0.00026187609126078064	0.0021515016201125 26	252
VTIP	-0.00003036303336313385 4	0.0023534495368726766	252
AGG	-0.00051567435107464 24	0.0041603638277181 91	252
TIP	-0.00031589055617145854	0.0050338361592590 6	252

Figure 12: 12M mu and sigma of the fixed income candidates

After reading key findings,

Business insights

- No bond fund made money in this period. Rising rates hurt every type. The goal is therefore the smallest loss with the lowest risk, not the highest gain.
- VTIP lost almost nothing. Its mu is about 6 times better than SHY's and 17 times better than BNDX's. The inflation protection offset most of the rate damage, which supports the argument for choosing it.
- SHY is the safest, but loses more. It has the lowest sigma (0.13%), but its mu is clearly worse than VTIP's.

3.4 Decision: VTIP

VTIP offers the best balance: the best mu of the group, very close to zero, with low risk (0.24%). That is about 9 times less volatile than ROK (2.11%) and 18 times less than BIDU (4.24%). It also gives a preview of Q4: VTIP has by far the best mu / sigma ratio of the six candidates.

3.5 Portfolio Risk Before and After the Change

First: her total portfolio value

Why: to calculate exactly how much 10% is.

```sql
SELECT SUM(h.quantity * h.value) as total_mv
FROM invest.account_dim a
JOIN invest.holdings_current h
ON a.account_id = h.account_id
WHERE a.client_id = 128;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_mv
1225775.8649120003

Figure 13: Total market value of Client *portfolio*

Her total portfolio value is $1,225,776, so 10% is about $122,578.

Second: Portfolio sigma BEFORE (current portfolio)

Why: on each day, her portfolio return is the weighted average of her holdings' returns that day. The sigma of those daily portfolio returns is her portfolio risk.

1. Subquery d is the daily returns, using the same LAG logic as Question 2.
2. Subquery w is the market value per ticker, the weights from Q1.
3. JOIN them on ticker, then GROUP BY date. SUM(ROR × mv) / SUM(mv) gives one portfolio return per day.
4. Outer query: AVG gives the portfolio mu and STD gives the portfolio sigma.

```
276  SELECT AVG(pd.port_ROR) as mu,
277         STD(pd.port_ROR) as sigma,
278         COUNT(*) as n_days
279  FROM (
280   SELECT d.price_date, SUM(d.ROR * w.mv)/SUM(w.mv) as port_ROR
281   FROM (
282    SELECT z.ticker, z.price_date, (z.p1 - z.p0)/z.p0 as ROR
283    FROM (
284     SELECT ticker, price_date, price as p1,
285           LAG(price, 1) OVER(
286            PARTITION BY ticker
287            ORDER BY price_date
288           ) as p0
289     FROM invest.michel_ward
290     WHERE price_date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
291    ) z
292    WHERE z.p0 IS NOT NULL
293   ) d
294   JOIN (
295    SELECT h.ticker, SUM(h.quantity * h.value) as mv
296    FROM invest.account_dim a
297    JOIN invest.holdings_current h
298    ON a.account_id = h.account_id
299    WHERE a.client_id = 128
300    GROUP BY h.ticker
301   ) w
302   ON d.ticker = w.ticker
303   GROUP BY d.price_date
304  ) pd;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mu	sigma	n_days
0.000027974085531447197	0.01400165186032426	252

Figure 14: Portfolio mu and sigma before the change

Her current sigma is about 1.40% and mu of 0.0028%

Third: Simulating the new portfolio

To measure how much risk VTIP would add, I compared Client portfolio sigma before and after the change. The decision was to move 10% of her portfolio, about $122,578, from ROK into VTIP. I chose ROK because it is her

largest position (about 21% of the portfolio), with high risk (sigma of 2.11%) and a negative average daily return. After the change, ROK falls to about 11% of the portfolio.

First, I needed VTIP's daily prices next to Client 3 holdings. My view only contains the securities she already owns, so VTIP's prices had to come from the pricing_daily_new table. I wrote two queries with the same structure and stacked them with UNION ALL. The first takes her holdings from my view. The second takes VTIP from pricing_daily_new, with the same filters my view already applies: Adjusted prices and the same 12 month window. UNION ALL requires both queries to have the same number of columns in the same order, which is why both return the ticker, the date, the current price (p1) and the previous day's price (p0). In both queries, LAG with PARTITION BY ticker places the previous day's price next to the current price, so the daily return can be calculated with (p1 − p0)/p0 without mixing prices from different securities.

Next, I built the new portfolio weights. I started from the same market value query as in Question 1 (quantity × value per ticker), then reduced ROK by $122,578 using a CASE statement. CASE works like an "if": when the ticker is ROK, it subtracts 122,578, and otherwise it subtracts 0, so all other holdings stay the same. I then used UNION ALL to add a new row for VTIP worth exactly $122,578. Her total portfolio value does not change. Money is only moved from ROK to VTIP, which simulates selling part of ROK and buying VTIP with the proceeds.

Finally, I joined the daily returns with the weights on ticker and calculated the portfolio return for each day as a weighted average, SUM(ROR × mv) / SUM(mv). The average of these daily portfolio returns gives the portfolio mu, and their standard deviation gives the portfolio sigma.

```sql
SELECT AVG(pd.port_ROR) as mu,
       STD(pd.port_ROR) as sigma,
       COUNT(*) as n_days
FROM (
 SELECT d.price_date, SUM(d.ROR * w.mv)/SUM(w.mv) as port_ROR
 FROM (
  SELECT z.ticker, z.price_date, (z.p1 - z.p0)/z.p0 as ROR
  FROM (
   SELECT ticker, price_date, price as p1,
          LAG(price, 1) OVER(
           PARTITION BY ticker
           ORDER BY price_date
          ) as p0
   FROM invest.michel_ward
   WHERE price_date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
   UNION ALL
   SELECT ticker, date, value as p1,
          LAG(value, 1) OVER(
           PARTITION BY ticker
           ORDER BY date
          ) as p0
   FROM invest.pricing_daily_new
   WHERE price_type = 'Adjusted'
   AND ticker = 'VTIP'
   AND date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
  ) z
  WHERE z.p0 IS NOT NULL
 ) d
 JOIN (
  SELECT h.ticker,
         SUM(h.quantity * h.value)
         - CASE WHEN h.ticker = 'ROK' THEN 122578 ELSE 0 END as mv
  FROM invest.account_dim a
  JOIN invest.holdings_current h
  ON a.account_id = h.account_id
  WHERE a.client_id = 128
  GROUP BY h.ticker
  UNION ALL
  SELECT 'VTIP', 122578
 ) w
 ON d.ticker = w.ticker
 GROUP BY d.price_date
) pd;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

mu	sigma	n_days
0.00007929661206526656	0.012525486739455296	252

Figure 15: Portfolio mu and sigma after moving 10% from ROK to VTIP

Results: portfolio risk before and after adding VTIP

Measure	Before	After (10% ROK → VTIP)	Change
Average daily return (mu)	0.0028%	0.0079%	+0.0051 percentage points
Daily risk (sigma)	1.400%	1.253%	−0.147 percentage points (about 10.5% less risk)
Trading days	252	252	Same period

Adding VTIP does not add risk to Client 3's portfolio. It reduces it. By moving only 10% of her money from ROK into VTIP, her daily portfolio sigma falls from 1.40% to 1.25%, which means about 10.5% less volatility. In simple terms, her portfolio's typical daily swing becomes smaller, so she would experience fewer sharp ups and downs in the value of her investments.

At the same time, her average daily return improves, from 0.0028% to 0.0079%, almost three times higher. Usually, lowering risk means giving up some return. In this case, she gives up nothing, because the money comes out of ROK, which had a negative average daily return over the last 12 months, and goes into VTIP, which stayed almost flat. I checked this result by hand: moving 10% from ROK (mu of −0.054%) into VTIP (mu of −0.003%) should raise portfolio mu by about 0.10 × 0.051% = 0.0051%, which matches the SQL result.

Question 4: Risk Adjusted Returns

4.1 Method and SQL Code

Step 1: start from my Q2 query

My Q2 code already calculates mu and sigma per ticker over the last 12 months. I will keep it exactly the same so Q2 and Q4 use the same data and window.

Step 2: add the risk adjusted column

In the outer SELECT, I will add:

```
AVG(o.ROR)/STD(o.ROR) as risk_adj
```

4.2 Results

```sql
SELECT o.ticker, o.major_asset_class,
       AVG(o.ROR) as mu,
       STD(o.ROR) as sigma,
       AVG(o.ROR)/STD(o.ROR) as risk_adj,
       COUNT(o.ROR) as n_days
FROM (
 SELECT z.ticker, z.major_asset_class,
        (z.p1 - z.p0)/z.p0 as ROR
 FROM (
  SELECT ticker, major_asset_class, price_date, price as p1,
         LAG(price, 1) OVER(
          PARTITION BY ticker
          ORDER BY price_date
         ) as p0
  FROM invest.michel_ward
  WHERE price_date >= DATE_SUB('2022-09-09', INTERVAL 12 MONTH)
 ) z
) o
WHERE o.ROR IS NOT NULL
GROUP BY o.ticker, o.major_asset_class
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ticker	major_asset_class	mu	sigma	risk_adj	n_days
ARB	alternatives	0.0001880738250165901	0.003990094643374493	0.04713517894340791	252
BCI	commodities	0.0009208222153526999	0.014678730333099809	0.06273173458853533	252
BIDU	equity	0.0001752242439155321	0.04247624560846277	0.004125229087587280 5	252
BIL	fixed_income	0.0000157150389826717 5	0.000118580160150265 79	0.13252671410426095	252
BNDX	fixed_income	-0.0005141802762166533	0.003726292513735451	-0.13798709422873764	252
CDNS	equity	0.0005172889863138617	0.024997443563370716	0.02069367553535979 3	252
CEG	equity	0.005142053261819878	0.03264409737273617	0.1575186228342291	161
CTA	alternatives	0.001093095316759 19	0.009256101599352083	0.11809456767800663	128
DBA	commodities	0.0004515499755440898	0.009755902331671966	0.04628479869854367	252
DBC	commodities	0.0012618309005981642	0.016260232639825532	0.07760226612672723	252
DIS	equity	-0.0017023671033584713	0.019747893239064 13	-0.08620499831298198	252
DJP	commodities	0.0010746830968422 22	0.016233167163001758	0.06620292183595657	252
FLSP	alternatives	0.0002251834320612296 5	0.008607266950123893	0.02616201325764485	252
GIGB	fixed income	-0.0006729189855169743	0.005448658225924101	-0.12350177926655442	252
GLDM	commodities	-0.0001482051413082971	0.008620259927985055	-0.01719265341723162 3	252
GOOGL	equity	-0.0008025805537272152	0.021452694892630023	-0.03741164258122825 5	252
IAUM	commodities	-0.0001480905042886996 7	0.008582079344920663	-0.01725578363201076 7	252
LBAY	alternatives	0.0008198413664641813	0.009042190163673166	0.09066844996888908	252
MARB	alternatives	0.0001129148382999403 4	0.002572550406795592 3	0.04389217719569925	252
MSVX	alternatives	-0.0003080103844002208 6	0.005065712555844975	-0.06080297312661947	252
OTIS	equity	-0.0006413008556868329	0.016422797450845208	-0.03904942855237053 5	252
RINF	alternatives	0.0005850210768678907	0.012269567696105453	0.04768065928301494	252
RLY	alternatives	0.0005430479237441 14	0.010416942673669724	0.05225187556711987 5	252
ROK	equity	-0.0005383799828615856	0.021079427038758494	-0.02554094158453513 2	252
SGOL	commodities	-0.0001513381123289644 6	0.008604148810403273	-0.01758896965449756	252
TSCO	equity	0.0002712006439428 17	0.020871363013411 62	0.01299391150297887 2	252
UNG	commodities	0.0029565260391413075	0.04679267081824431	0.06318352826290 41	252

Figure 16: 12M risk adjusted return (mu / sigma) per security

4.3 Best Security and Why

Looking only at the number, CEG has the highest risk adjusted return (0.158), followed by BIL (0.133) and CTA (0.118). However, each of these needs a closer look:

- CEG and CTA are based on only 161 and 128 trading days, because they started trading in 2022. Their scores do not cover a full year, so they are less reliable than the others.
- BIL ranks high only because its risk is almost zero (sigma of 0.01%). Its average return is also almost zero, so dividing a tiny return by a tiny risk gives a high ratio. BIL is a safe place to keep cash, but it does not grow her wealth.

For these reasons, I consider LBAY (Leatherback Long/Short Alternative Yield ETF) the best security in Client ; portfolio. It has the highest risk adjusted return (0.091) among securities with a full year of data and a meaningful return. It earned a solid average daily return (0.082%) with low risk (sigma of 0.90%, lower than any of her stocks), and in Question 1 it returned +21.7%

over 12 months. LBAY gives her strong return without large swings, which is exactly what a long/short alternative fund is designed to do.

4.4 Recommendations

1. Increase LBAY. It is her best full year risk adjusted holding, so part of the money taken from her weakest positions should go there.

2. Reduce ROK, sell DIS, and review GOOGL and OTIS. All four have negative risk adjusted returns, so she is taking risk without being rewarded.

3. Replace BNDX and GIGB with short duration bonds such as VTIP. They had the worst risk adjusted returns in the portfolio, as they lost value steadily while rates rose.

4. Keep only one of the three gold ETFs. GLDM, IAUM and SGOL all have the same risk adjusted return of about −0.017, so holding all three adds no diversification.

5. Monitor CEG and CTA. Their early results are strong, but they have less than a year of data, so they need more history before increasing them.

Appendix:

Claude conversation used for this assignment:
https://claude.ai/share/8c63a57f-161b-4fa4-8efe-24bfc0299287

Step 1 Details (Client Confirmation and Table Joins)

Confirm the client and list her accounts

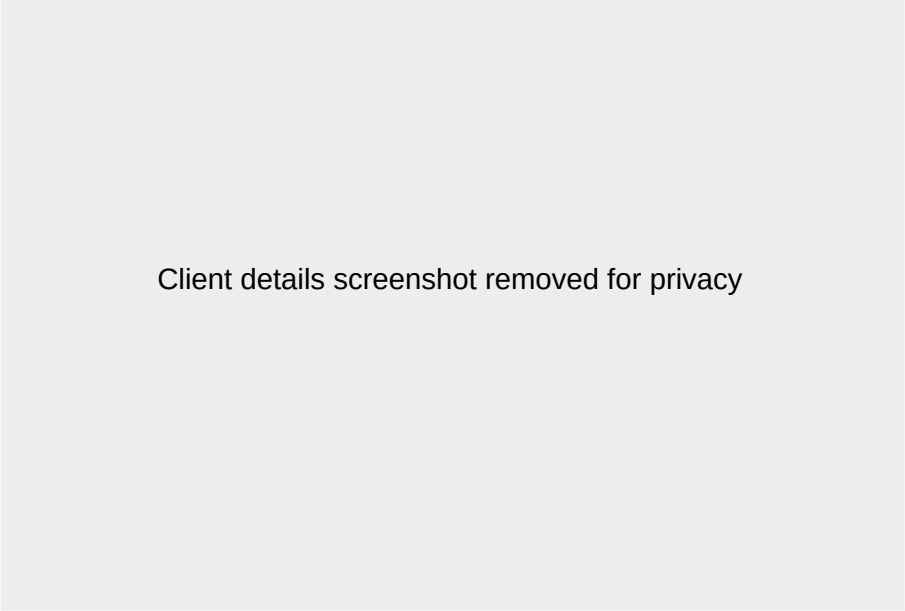


Figure 17: SQL code to confirm the client and list her accounts

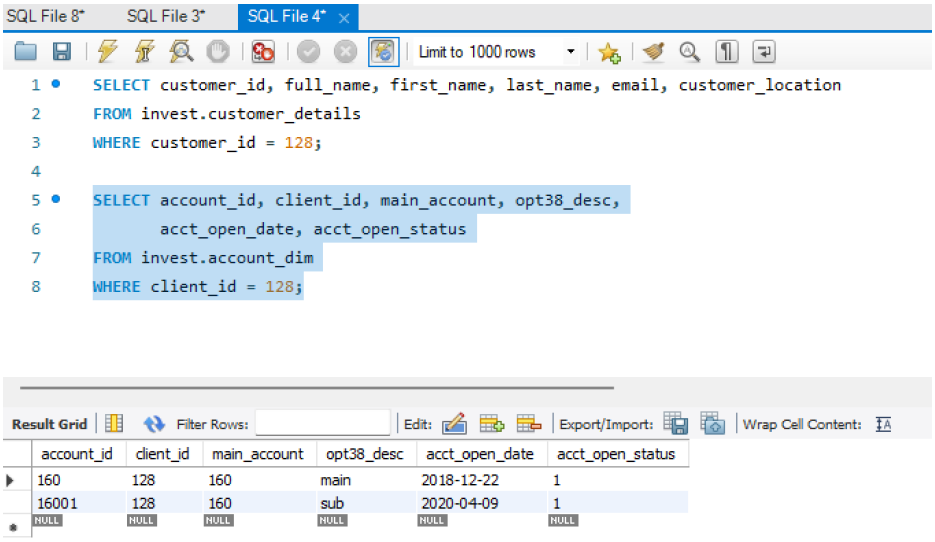


Figure 18: Client details and accounts for customer #128

Key Findings:

1. **Client confirmed.** Customer #128 is Client #128 based at Hult's Dubai campus ([email removed]).
2. **Two accounts.** Client holds a **main account (160)** and a **sub account (16001)**.

3. The main account was opened on **22 Dec 2018** and the sub account on **9 Apr 2020**.
4. acct_open_status = 1 for both

Business Insights:

- Long lasting client. She opened her main account in December 2018, almost 4 years before the last date in the data (2022-09-09)

Questions:

- What assets does she have in each account?
- How much in value does she have?
- How diversified is her portfolio?
- Is she a risk taker or more conservative?

Full join across all five tables

Join	Connecting key	Why it's needed
customer_details → account_dim	customer_id = client_id	Finds all of Client ; accounts (160 and 16001)
account_dim → holdings_current	account_id	Finds what each account holds
holdings_current → security_masterlist	ticker	Adds security name, type and asset classes
security_masterlist → pricing_daily_new	ticker	Adds the daily price history